

Krish Lakshmi Narayanan

+1 (226) 995-0215 | krish.narayanan1201@gmail.com | linkedin.com/in/krish1201 | krishnara1201.github.io

Technical Skills

Languages: Python, SQL, R, SAS, Go, TypeScript/JavaScript

Machine Learning: Classification, regression, clustering, feature engineering, model evaluation, XGBoost, LightGBM, logistic regression, neural networks

Experimentation & Statistics: A/B tests, Bayesian & frequentist methodologies, experimentation frameworks, hypothesis testing, causal inference, Sampling

Data Engineering & MLOps: ETL pipelines, Databricks, Airflow, dbt, Fivetran, PostgreSQL, Snowflake, BigQuery, Apache Spark, AWS, GCP

Analytics & Visualization: Power BI, Looker, Tableau, forecasting, pricing analytics, cohort analysis

Software & Infrastructure: REST APIs, distributed systems, microservices, scalable infrastructure, version-controlled modular codebases

Professional Experience

Senior Analyst – Advanced Analytics & Optimization – CIBC – Toronto, Canada – (August 2025 – Present)

- Trained and deployed **XGBoost** classification model achieving **0.86 AUC** to score behavioral risk and surface compliance anomalies across **500,000+** employee records, reducing manual review volume by **30%** and identifying high-risk cases **2x** faster than rule-based approaches
- Engineered **ETL pipelines** processing **3B+** records in **Python and Databricks**—decreasing end-to-end data latency by **35%** through optimized partitioning and parallel execution strategies that scale to multi-terabyte volumes
- Architected production analytics systems serving **10,000+ users** across distributed infrastructure, integrating **Python** services with cloud-native workflows to sustain high availability and eliminate single points of failure
- Built self-service analytics via bespoke **Power BI dashboards** embedded with **forecasting models**, reducing ad-hoc reporting requests by **40%** and enabling non-technical stakeholders to close reviews faster

Data Intern – 1Password – Toronto, Canada – (May 2024- August 2024)

- Shipped **LightGBM** and **logistic regression** models with precision/recall evaluated across user segments, combined with **k-means clustering** for cohort analysis—increasing website engagement by **5%** through targeted funnel optimizations
- Stood up an experimentation framework and executed **5+ A/B tests** using Bayesian and frequentist methodologies, boosting experimentation throughput by **40%** and informing high-impact product launches across activation and retention funnels
- Translated complex multi-source analyses into **Looker dashboards** tracking conversion, activation, and user lifecycle metrics for senior stakeholders, accelerating decision speed by **25%**
- Constructed **ETL pipelines** ingesting **2B+** clickstream and Google analytics records into AWS, maintaining **100% data integrity** across downstream experimentation and reporting layers

Data Analyst – Super.com – Toronto, Canada - (January 2023 – April 2023)

- Developed pricing and revenue optimization models using **regression techniques** across **1M+ transaction** records, surfacing actionable spend patterns that contributed to **5%+ revenue growth** in key e-commerce segments
- Designed and executed **5+ A/B experiments**—sample size calculations, experiment design, and post-hoc statistical analysis via Bayesian and frequentist methodologies—generating customer behavior insights that produced a **5% net revenue lift** in key e-commerce segments
- Automated **15+ production datasets** and reporting workflows using **Airflow, FiveTran** and **dbt**, raising data reliability by **45%** and enabling scalable analytics infrastructure that freed analyst capacity for higher-leverage insight work

Small Business Data Analyst – Scotiabank – Toronto, Canada - (January 2022 – August 2022)

- Pioneered **statistical and regression-based** scoring models—linear regression—across **600+ advisors** to evaluate training certification, performance, and behavioral risk, directly informing compensation, promotion, and workforce planning decisions
- Applied predictive modeling techniques—**classification** and **probability-of-event modeling**—across **100,000+** customer and advisor records to assess churn and onboarding quality, combining approaches to sharpen retention strategy for financial products

Projects

ModuRisk – Ops Platform for Risk Analytics – Python (Flask), React, MongoDB, Docker, CI/CD

- Built a full-stack risk analytics platform that functions as an ops/fraud investigations hub, enabling real-time scoring and monitoring
- Designed backend services and data models for large-scale, low-latency analytics, improving decision turnaround by **30%**.
- Set up **CI/CD** and **containerized deployments**, reducing release cycles by **60%** and increasing reliability for new features.

QuantTrade – Algorithmic Trading & Backtesting Platform - Python (FASTAPI, Numpy, Sci-kit), JavaScript, dbt, PostgreSQL

- Built backend services enabling strategy back-testing across **1M+** historical market records
- Architected modular pipelines supporting **20+** trading strategies, improving computational throughput by **40%**
- Implemented clean API layers and relational schemas emphasizing correctness and extensibility

Reddit Sentiment Analysis Data Pipeline - Python, Apache Airflow, dbt, PostgreSQL, Docker, VADER, TextBlob

- Developed ingestion and transformation workflows processing **500K+** records, converting raw data into structured outputs
- Orchestrated ETL workflows with failure recovery and monitoring, improving reliability by **27%**

Education

Honors Bachelor of Mathematics (Co-op) – University of Waterloo

Double Major: Computational Mathematics & Statistics | **Minor:** Economics

- **Dean's Honors** in Computational Mathematics and Statistics; **Term Honors distinction** in 8 academic terms
- **Coursework:** Data Structures & Algorithms, Neural Networks, Advanced Regression, Classification